

in the face of new evidence. A classic experiment by Ward Edwards¹ in 1968 eloquently illustrated the technical side of conservatism bias. Edwards presented subjects with two urns—one containing three blue balls and seven red balls, the other containing seven blue balls and three red ones. Subjects were given this information and then told that someone had drawn randomly 12 times from one of the urns, with the ball after each draw restored to the urn in order to maintain the same probability ratio. Subjects were told that this draw yielded eight reds and four blues. They were then asked, “What is the probability that the draw was made from the first urn?” While the correct answer is 0.97, most people estimate a number around 0.7. They apparently overweight the base rate of 0.5—the random likelihood of drawing from one of two urns as opposed to the other—relative to the “new” information regarding the produced ratio of reds to blues.

Professor David Hirshleifer of Ohio State University² noted that one explanation for conservatism is that processing new information and updating beliefs is cognitively costly. He noted that information that is presented in a cognitively costly form, such as information that is abstract and statistical, is weighted less. Furthermore, people may overreact to information that is easily processed, such as scenarios and concrete examples. The argument for costly processing can be extended to explain base rate underweighting. If an individual underweights new information received about population frequencies (base rates), then base rate underweighting is really a form of conservatism. Indeed, base rates are underweighted less when they are presented in more salient form or in a fashion that emphasizes their causal relation to the decision problem. This argument for costly processing of new information does not suggest that an individual will underweight his or her preexisting internalized prior belief. If base rate underweighting is a consequence of the use of the representativeness heuristic, there should be underweighting of priors.

Portions of this analysis resonate interestingly with Edwards’s experiment. For example, perhaps people overweight the base rate probability of drawing randomly from one of two urns, relative to the sample data probability of drawing a specific combination of items, because the former quantity is simply easier to compute.

PRACTICAL APPLICATION

James Montier is author of the 2002 book *Behavioural Finance: Insights into Irrational Minds and Markets*³ and analyst for DKW in London. Montier has done some exceptional work in the behavioral finance field. Although Montier primarily studied the stock market in general, concentrating on the